idea, Einstein could only point to the photoelectric effect itself, along with a strange quirk of the "black-body radiation law" discovered by German physicist Max Planck five years earlier. For nearly two decades, almost nobody other than Einstein believed in photons. Even Planck himself didn't think his work suggested that light was made of particles (though, years later, Planck's work was hailed as the start of the quantum revolution). Only when Arthur Compton actually caught photons in the act of bouncing off of electrons, in 1923, did the physics community finally come around to Einstein's way of thinking—and even then there were a few holdouts.

But Einstein was accustomed to isolation. He had changed the world in 1905 working alone in a Swiss patent office and continued that habit for the rest of his life. Einstein once said he went through life as a "one-horse cart"; he rarely collaborated with other physicists and almost never took on students of his own. He was eternally suspicious of the status quo, both scientifically and elsewhere; he characterized common sense as the collection of prejudices accumulated by the age of eighteen. So when Heisenberg's astonishing new theory arrived on the scene in 1925, Einstein was unsurprisingly skeptical. "Heisenberg has laid a big quantum egg," he wrote to his friend Paul Ehrenfest shortly after Heisenberg's ideas were first published. "In Göttingen they believe in it. I don't." Presented with the opportunity to interrogate Heisenberg at close quarters, Einstein pounced.

Safely ensconced in his apartment, Einstein finally asked Heisenberg what he really wanted to know. "You assume the existence of electrons inside the atom, and you are probably quite right to do so. But you refuse to consider their orbits.… I should very much like to hear more about your reasons for making such strange assumptions."

"We cannot observe electron orbits inside the atom," replied Heisenberg. He pointed out that only the spectrum of light from an atom is really observable and concluded with a rather Machian statement. "Since a good theory must be based on directly observable magnitudes, I thought it more fitting to restrict myself to these."

In Heisenberg's later retelling of this encounter, Einstein was shocked at